FIRSTLIGHT

TECHNICAL BLUEPRINT V2.0

Autonomous Geo-AI & Tactical Logistics Engine for Disaster Response

Product Scope: FirstLight ingests pre- and post-event satellite, SAR, and drone imagery, detects damage footprints and flood extents, and fuses those outputs with road networks, population grids, and critical facility data. It resolves actionable tactical questions for field logistics within the critical first six hours post-disaster, outputting interactive maps, APIs, citation-grounded sitreps, and offline field bundles.

< 30 min	100 km²	IoU ≥ 0.80	0.5m / px
TARGET PROCESSING TIME	AOI TILE PROCESSING SCALE	SAR FLOOD ACCURACY	MAX SPATIAL RESOLUTION

SYSTEM ARCHITECTURE CORE LAYERS

Layer	Technology Stack	Operational Focus
1. Ingestion Sentinel	`pystac-client`, Copernicus API, Microsoft Planetary Computer, `rasterio`	Continuous STAC polling, auto-triggering on AOI bounds & low cloud-cover thresholds.
2. Preprocessing	ASF HYP3 (RTC), ESA SNAP, `pyroSAR`, `rioxarray`	Sentinel-1 GRD to radiometrically terrain-corrected gamma0 (dB scale), speckle filtering.
3. AI Ingestion Mesh	PyTorch, `torchgeo`, TerraTorch, ONNX Runtime (FP16/INT8)	Siamese building damage classification, physics-driven SAR flood extent segmentation.
4. Spatial Graph Engine	PostGIS, `pgRouting`, H3 Spatial Index, `pgvector`	Dynamic road blockage graph, hospital isolation deltas, neighborhood prioritization.
5. Grounded LLM Layer	Claude API / Ollama (Local), Outlines, Deterministic Tool Mesh	Citation-backed situation report drafting with cryptographic tool assertion verification.

Decision Layer Wedge

Unlike standard raw segmentation services, FirstLight fuses geospatial polygons with dynamic graph networks to calculate active routing isolations and population risk indices instantly.

Air-Gapped Operation

Built for disconnected field deployments via INT8 quantized ONNX pipelines, local SpatiaLite engine fallback, and Ollama-based local report generation.

MULTI-MODEL AI ARCHITECTURE

1. Building Damage Assessment (Optical Pre/Post)

Uses a dual-stage Siamese UNet architecture with ResNet-50 backbone. Stage A localizes building footprints (skipped if Google/MS Open Buildings exist). Stage B classifies damage into 4 major classes: *No Damage, Minor Damage, Major Damage, Destroyed*.

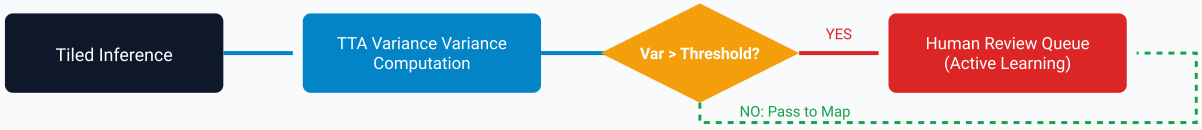
```
import torch
import segmentation_models_pytorch as smp

# Siamese Pre/Post Optical Fusion Architecture (6 Channels Input)
model = smp.Unet(
    encoder_name="resnet50",
    encoder_weights="imagenet",
    in_channels=6,
    classes=5
)

def damage_loss(logits, target, class_weights):
    ce = torch.nn.functional.cross_entropy(logits, target, weight=class_weights)
    dice = smp.losses.DiceLoss(mode="multiclass")(logits, target)
    return ce + dice
```

ACTIVE LEARNING & EPISTEMIC UNCERTAINTY PIPELINE

To eliminate model hallucinations in high-stakes life-and-safety environments, FirstLight uses Test-Time Augmentation (TTA) to derive epistemic uncertainty metrics across sliding-window inferences.



Uncertainty Routing Logic

High-variance spatial tiles are automatically flagged and queued for human-in-the-loop validation or pushed to secondary high-resolution drone tasking workflows.

SPATIAL DECISION LAYER & NEED SCORING

Raw building damage polygons are insufficient for rapid dispatchers. FirstLight aggregates data into standard **Uber H3 Hexagonal Indexing (Resolution 8–9)** and derives an objective, transparent score for prioritization.

Hexagonal Zone Need Equation

$$\text{Need}(z) = w_1 \cdot D(z) + w_2 \cdot P_{exp}(z) + w_3 \cdot \text{Iso}(z) + w_4 \cdot \text{Crit}(z)$$

- **D(z):** Weighted damage ratio = $(1.0 \cdot N_{destroyed} + 0.6 \cdot N_{major} + 0.2 \cdot N_{minor}) / N_{total}$
- **$P_{exp}(z)$:** Affected population normalized against peak zone count from WorldPop datasets.
- **Iso(z):** Share of population whose travel time to nearest hospital increases by > 30 mins post-event.
- **Crit(z):** Count of impacted critical infrastructure assets (hospitals, power nodes, water facilities).

PASSABLE ROAD NETWORK & ISOLATION ANALYSIS

The road network graph is constructed from OpenStreetMap via `osmnx` and stored in PostGIS. Roads overlapping flood extent masks or situated within destroyed building debris buffers are dynamically penalized or flagged as blocked.

Analysis Type	Spatial Engine Tool	Actionable Output for Responders
Isochrone Delta	`pgRouting` (pgr_drivingDistance)	Identifies communities cut off from emergency healthcare within 60 minutes.
Passable Route Engine	`pgRouting` (pgr_dijkstra)	Provides optimized dispatch trajectories avoiding confirmed flooded/blocked links.
Debris Proxy Buffers	PostGIS `ST_Buffer` + `ST_Intersects`	Flags narrow urban corridors susceptible to masonry blockages.
Facility Isolation Index	H3 Multi-Ring Traversals	Ranks hospitals operating on emergency power or disconnected supply routes.

Logistics Routing Confidence

Routes generated carry visual uncertainty tags. If a path relies on road segments with unverified optical imagery, the system explicitly advises field units to execute physical reconnaissance.

GROUNDED LLM SITUATION REPORT ENGINE

To eliminate model hallucinations, the LLM layer uses strict, typed tool-calling over PostGIS. Every numeric claim in the text must be bound to a verified database hash.

Tool Command	Input Parameters	Returned Payload
`get_event_summary()`	`event_id: str`	AOI size, date/time, total damage count breakdown, flood area.
`top_zones()`	`event_id: str, limit: int`	Ranked H3 zones, need scores, isolated population totals.
`isolated_facilities()`	`event_id: str, facility_type`	Hospitals/shelters with baseline vs post-event travel times.
`generate_passable_route()`	`origin_coord, dest_coord`	Routable coordinates, estimated travel time, obstacle warnings.

8-WEEK ENGINEERING IMPLEMENTATION ROADMAP

Phase	Core Engineering Focus	Exit Milestone
Weeks 1–2	STAC ingestion engine, DVC pipeline, SAR baseline script.	Reproducible data pipeline on 2 test events.
Weeks 3–4	Siamese UNet damage model, flood segmentation, H3 scoring.	Leave-one-event-out benchmark validation.
Weeks 5–6	FastAPI, pgRouting engine, MapLibre UI with before/after swipe.	Click-through interactive demo platform.
Weeks 7–8	ONNX INT8 optimization, offline Docker container, LLM tools.	Fully functional air-gapped field deployment.

Data Licensing & Ethics Guardrails

Prototype on public benchmarks (xBD, Sen1Floods11), then transition to open commercial datasets (OpenStreetMap, Copernicus) and proprietary pilot labels. Operational safety features include mandatory human-in-the-loop review queues prior to alert distribution.